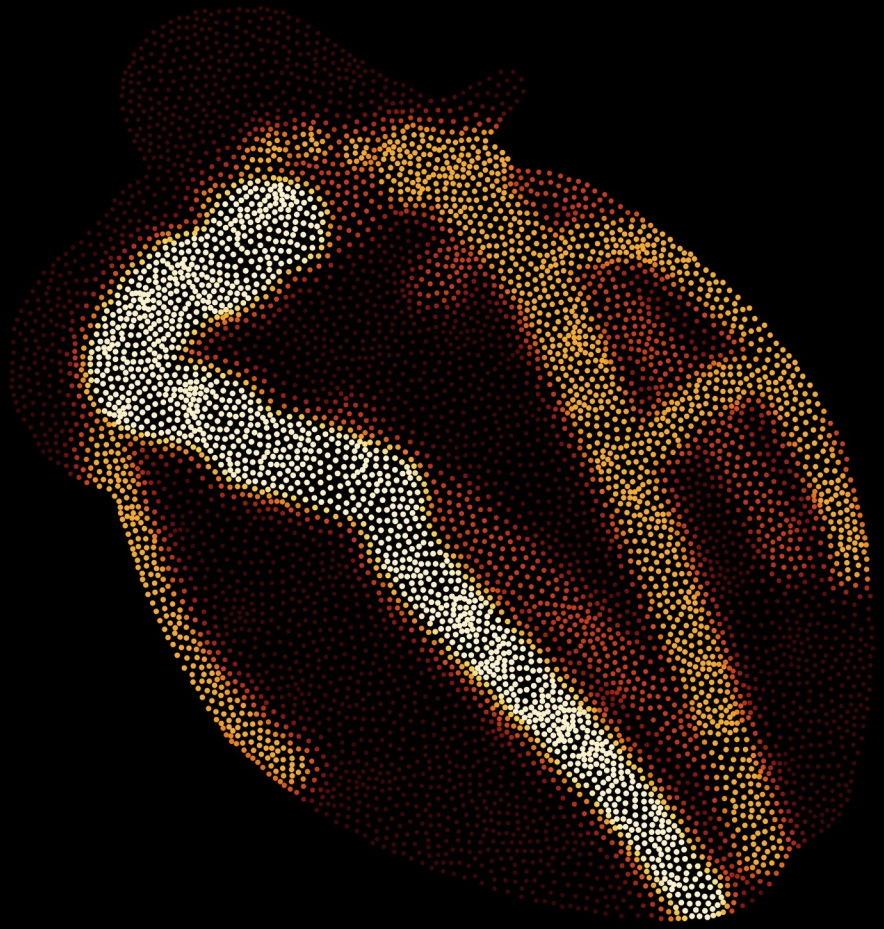
ACC CARDIO SIM CREATOR COMPETITION

CoroSim

A closed-loop cardiovascular simulator for learning cardiology by doing

corosim.com



Heart: weighted Voronoi stipple of CoroSim's own coronary tree

One simulated patient, six ways to learn from it

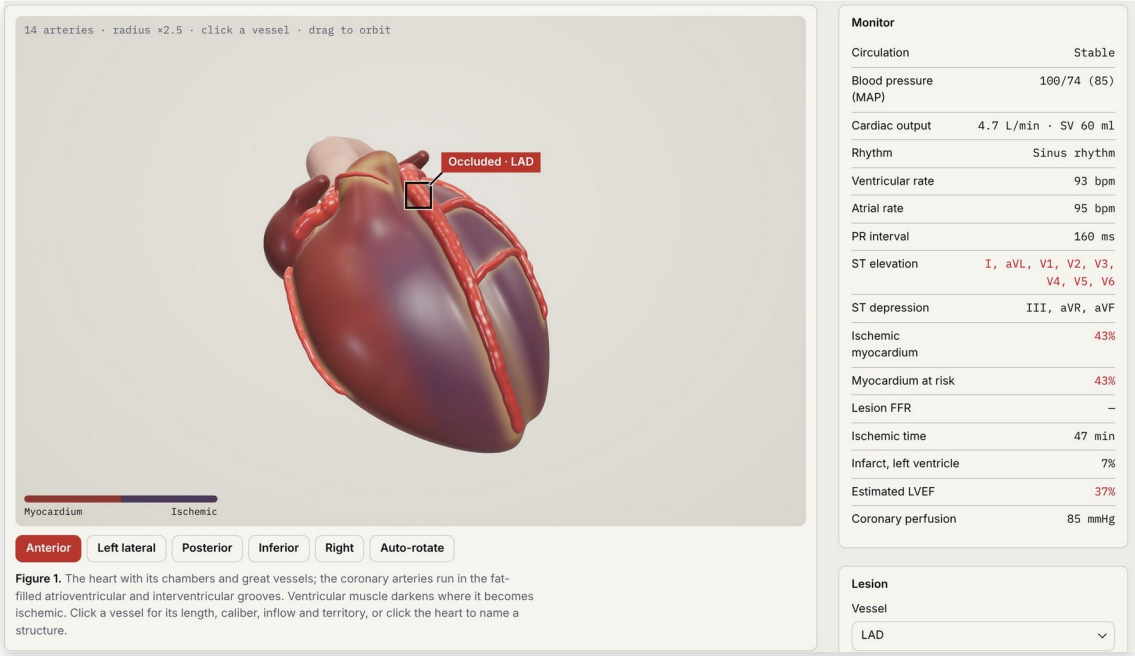


Figure 1. The heart with its chambers and great vessels; the coronary arteries run in the fat-filled atrioventricular and interventricular grooves. Ventricular muscle darkens where it becomes ischemic. Click a vessel for its length, caliber, inflow and territory, or click the heart to name a structure.

Proximal LAD occlusion, 47 clinical minutes in: anterior ST elevation, 43% of myocardium ischemic, LVEF down to 37%.

6

linked pages: simulator, daily case, EKG, cath lab, structural, echo

50+

EKG presets, plus custom tracings built on request

$0

free in any browser: no install, account or license

WHAT THIS SUBMISSION ANSWERS

The five competition criteria

01	**Problem solving and efficacy**	What trainees struggle with, and how CoroSim addresses it
02	**Skillsets targeted**	Six skill domains across the site
03	**Intended audience**	From medical students to fellows and CCU teams
04	**Assessment metrics and competency evaluation**	Proposed framework (draft)
05	**Significance and seamless integration**	Where CoroSim fits in existing programs

ECGs stress trainees, and practice is scarce

83% · 61%

of medical students and EM residents said reading ECGs in clinical settings makes them anxious

El-Baba et al., Int J Emerg Med 2023

<10%

of 165 residents were fully confident in their ECG diagnoses; none correctly identified LBBB

Liang et al., Adv Med Educ Pract 2025

71%

of 420 SCAI members felt their simulation training was insufficient; lack of access was the top barrier

Lee et al., JSCAI 2025

P = 0.45

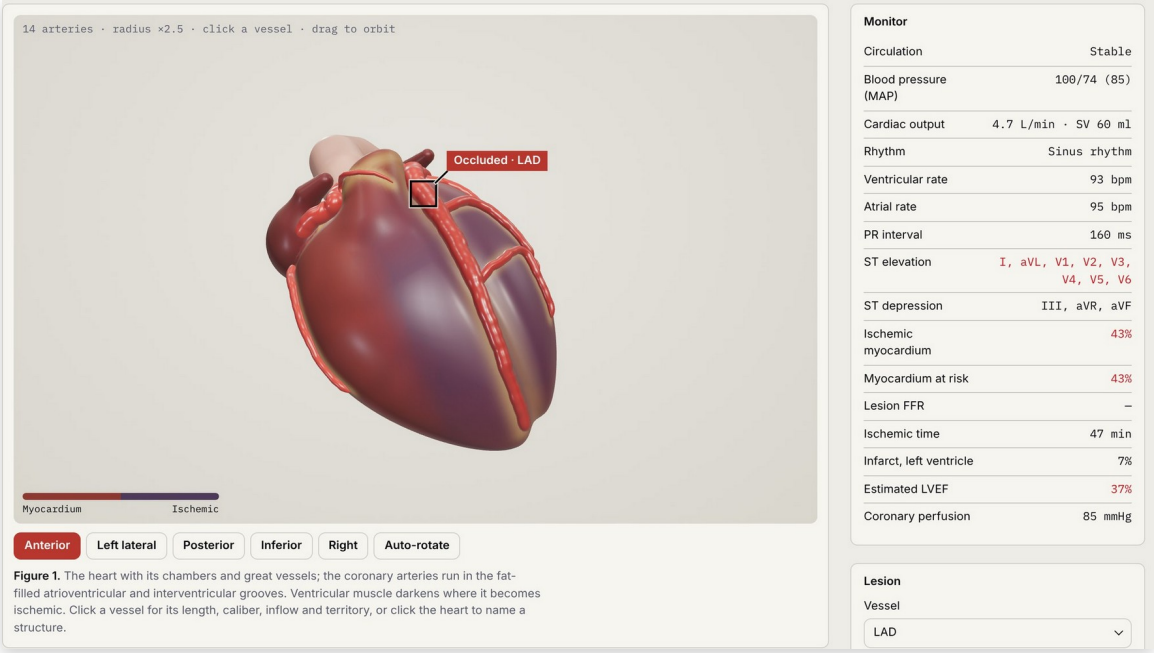
A 14-week lecture series raised residents' ECG confidence (2.8 to 3.5 of 5) but not their knowledge scores

Esomonye et al., JACC Case Rep 2026

Where learning breaks down, and what changes

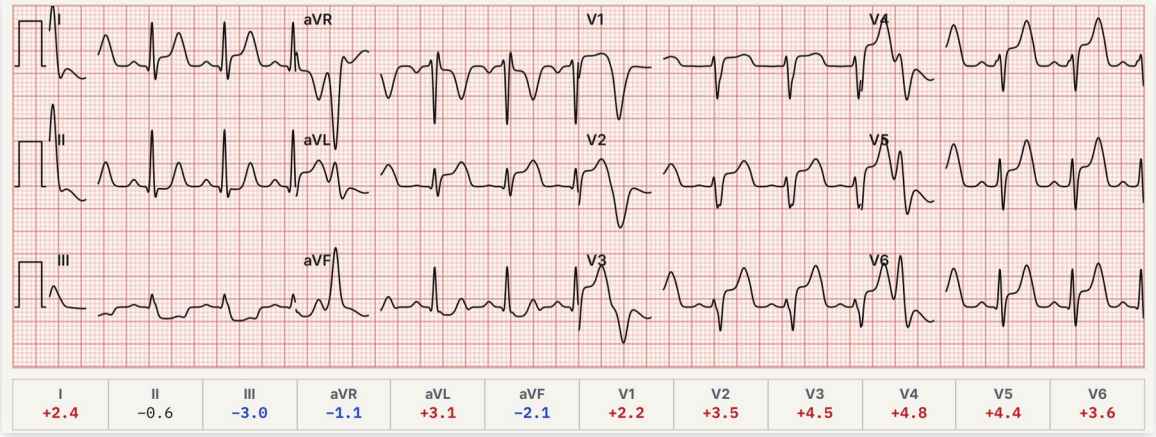
THE GAP	IN COROSIM
ECGs memorized as patterns, cut off from anatomy	Every tracing comes from a lesion in a 3D coronary tree; leads drawn as axes show why ST rises or falls
Physiology taught as static curves	A closed loop: lesion → ischemia → ECG → pressure → reflexes, beat by beat
STEMI and shock decisions rarely practiced	Treat while muscle dies; every drug and device has its real cost, and warnings name the harm
Simulation is scarce, costly and tied to a lab	Free, in any browser, any time; one link opens a specific case

Block an artery, watch the patient respond



- Place a lesion on any of 14 arteries, across nine anatomic variants
- Ischemia, the 12-lead, rhythm, AV block and blood pressure all follow from the physics
- Time is muscle: the infarct grows in clinical minutes and lowers LVEF until reperfusion

Read the 12-lead, then treat the patient



- ST shifts come from each ischemic bed, so the leads point to the artery
- Every drug and device has its real cost; warnings name the harm
- PCI, lysis and CABG arrive after their real delays

Management

Support | Reperfuse | **Assessment**

Acute occlusion, ischemic for 76 min

Infarct 14% of the left ventricle

What would the guidelines do?

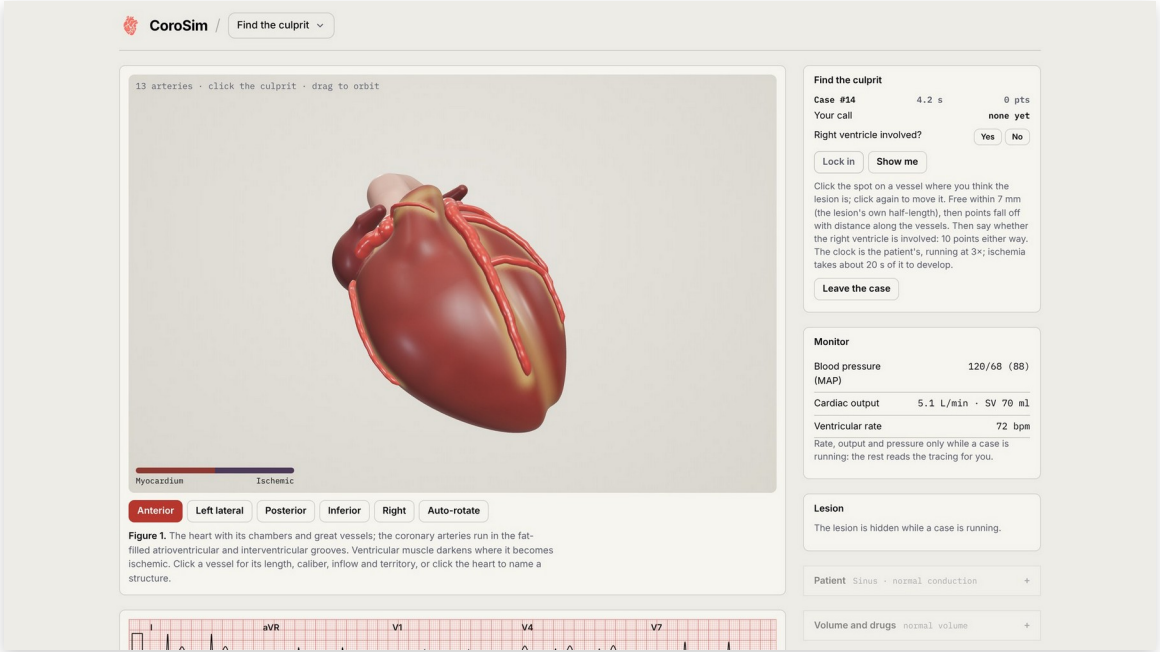
Reset treatment

1 **Reperfuse now: primary PCI**

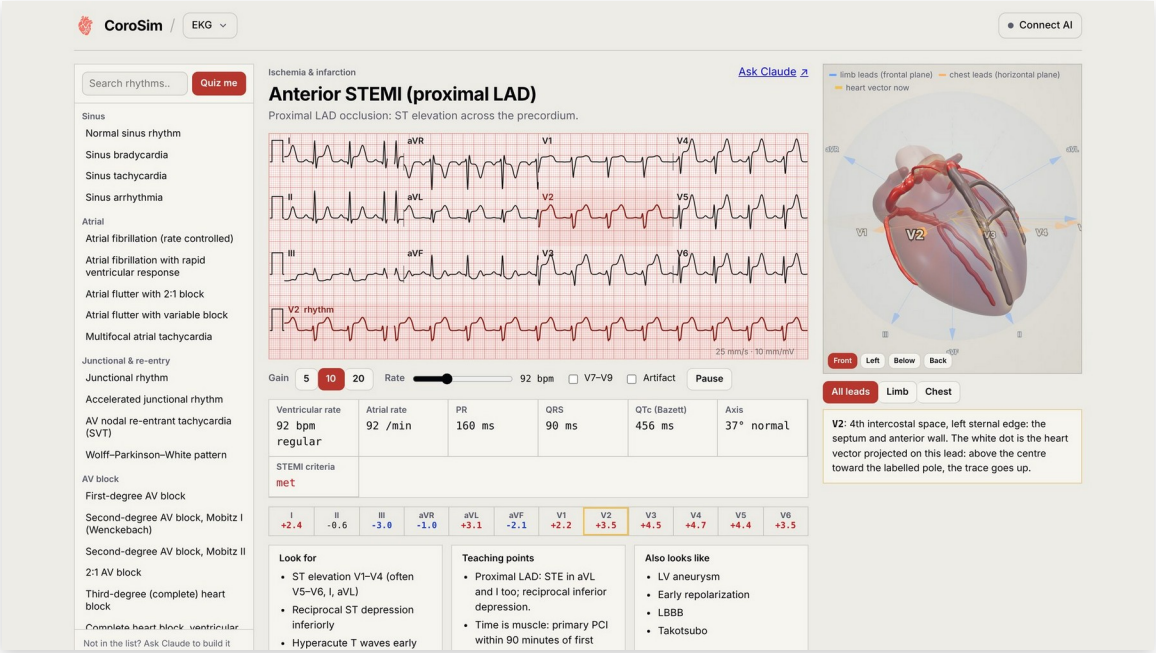
Primary PCI is preferred when first device time is 90 min or less; a fibrinolytic is the alternative if PCI cannot be done within 120 min of first contact, and works best in the first 3 h. CABG is for those who are not PCI candidates.

A daily hidden-lesion case, the same for everyone

- The lesion is hidden: read the tracing and monitor, then click where it is
- Points for accuracy along the vessel tree and for speed, then the RV call
- Then manage the case: points for muscle saved and time out of shock, minus harm



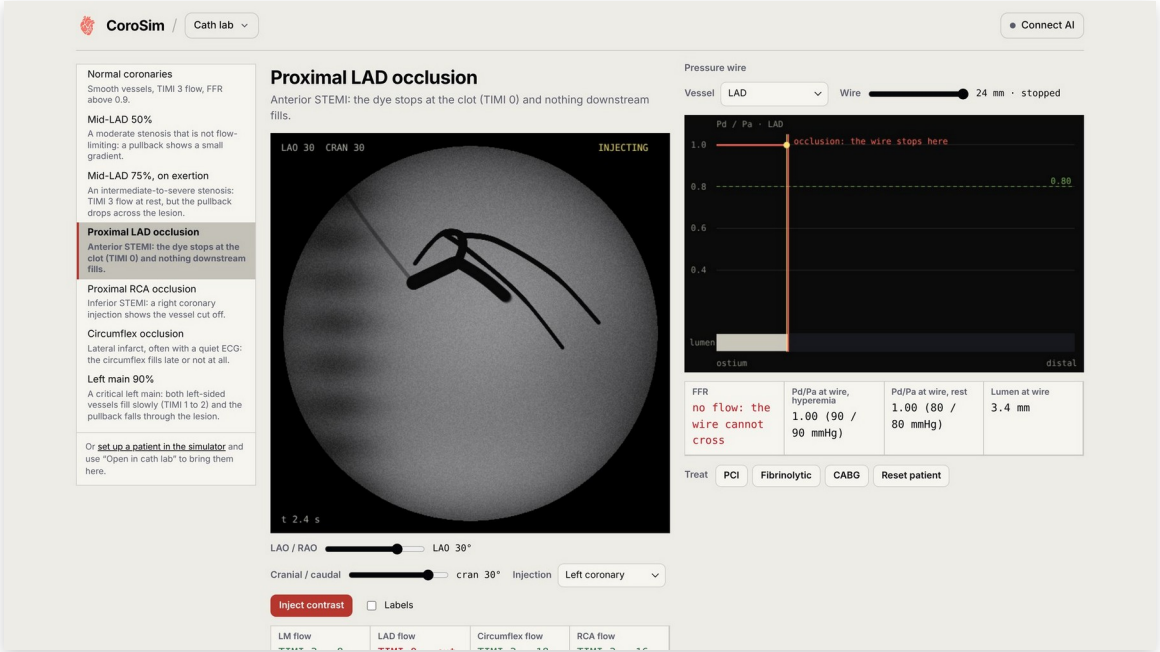
50+ tracings, each drawn through a 3D heart



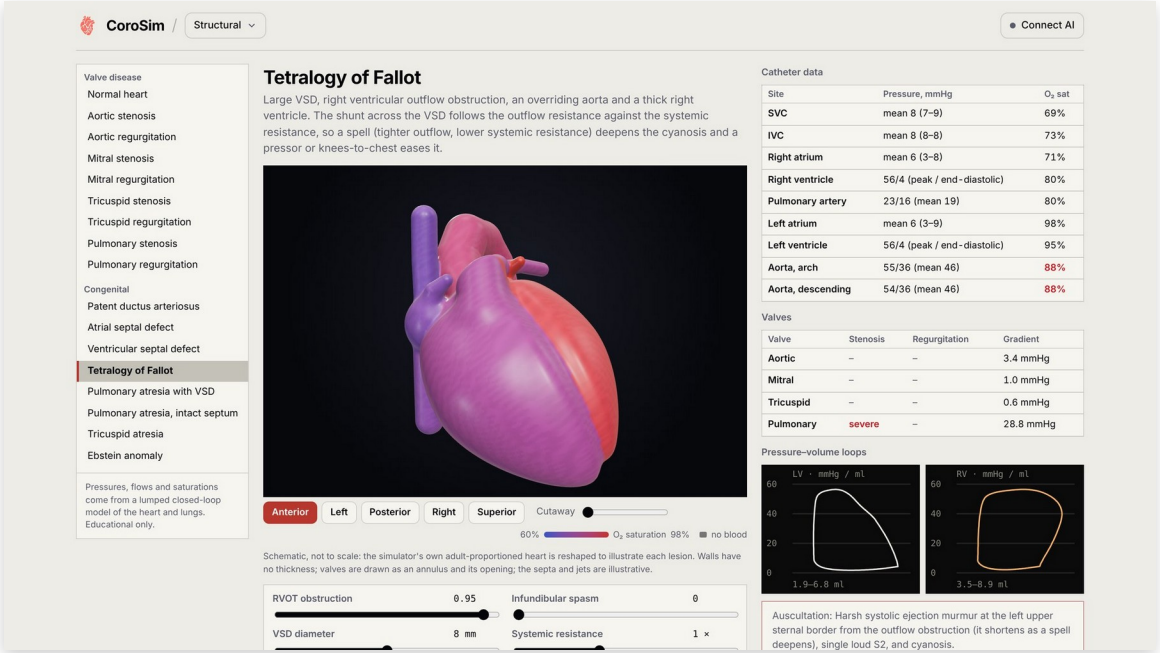
- Rhythms, blocks, ischemia patterns and STEMI mimics, with teaching points
- Each lead is an axis through the heart, so learners see why it goes up or down
- Intervals, axis and ST measured from the waveform; Quiz me hides the name

A virtual cath lab on the same patient

- Angiography from any C-arm angle; dye moves at each vessel's solved flow (TIMI 0 to 3)
- Pressure-wire pullback at rest and hyperemia reads Pd/Pa and FFR
- PCI, lysis and CABG show the vessel before and after



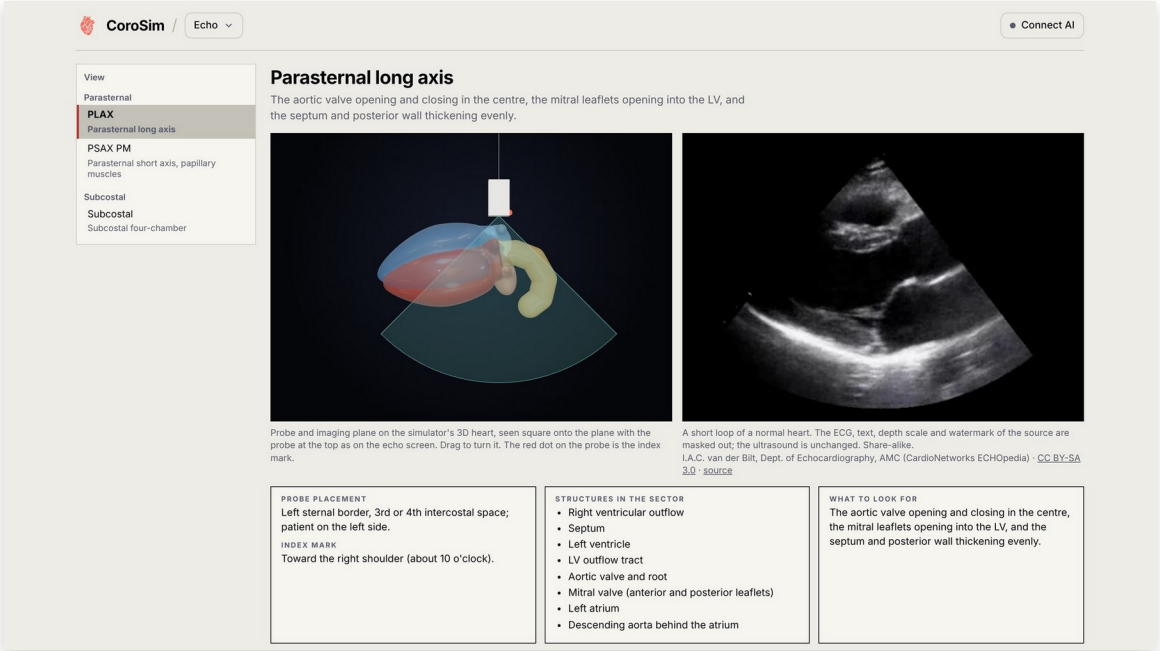
Valve and congenital lesions on a closed-loop heart



- 8 valve lesions and 8 congenital defects, each with severity sliders
- Cath pressures, saturations, gradients, Qp:Qs and PV loops update live
- Auscultation findings and oxygen saturation mapped onto the 3D heart

Echo atlas: from probe position to the loop

- Probe and imaging plane on the 3D heart beside a real TTE loop
- Parasternal long axis, short axis and subcostal views
- Probe placement, index mark, structures in the sector and what to look for



Six skill domains, practiced across the site

ECG interpretation

Localize the culprit from ST changes; 50+ rhythms and mimics

EKG · SIM · GAME

Coronary anatomy

Dominance, variants, and which bed each artery feeds

SIM · GAME · CATH

Ischemic physiology

Supply vs demand, autoregulation, reflexes, perfusion pressure

SIM · CATH

STEMI and shock care

Reperfusion choice and timing; drug and device trade-offs

SIM · GAME

Invasive and imaging reads

Angiographic views, TIMI flow, FFR, echo windows

CATH · ECHO

Valve and congenital hemodynamics

Gradients, shunts, saturations and PV loops

STRUCTURAL

Built for residents and fellows, open to every level

Medical students		Cardiovascular physiology labs; clerkship ECG and coronary anatomy
IM and EM residents	PRIMARY	ECG conference, STEMI and cardiogenic shock care, CCU preparation
Cardiology fellows	PRIMARY	Culprit localization, cath lab orientation (views, TIMI, FFR), echo windows, structural hemodynamics
Nurses, APPs, CCU and cath lab teams		Rhythm recognition and what each support drug or device does

Proposed assessment framework (to finalize)

KIRKPATRICK

1 · Reaction

- *Usability (SUS)*
- *Perceived realism*
- *ECG and STEMI confidence (Likert)*

KIRKPATRICK

2 · Learning

- *Pre/post culprit-localization test*
- *Rhythm ID accuracy*
- *Time to correct call*

KIRKPATRICK

3 · Behavior

- *Score trend across daily cases*
- *Guideline-concordant steps*
- *Harmful actions avoided*

KIRKPATRICK

4 · Results

- *Accuracy on real ECGs*
- *Milestone ratings*
- *Long-term follow-up*

Already measured in-app: culprit-call error (mm along the vessel), time to call, RV call, infarct vs. immediate-PCI and no-reperfusion baselines, time in shock, harmful actions by name, and EKG quiz answers.

Why CoroSim matters in the wider training picture

Cardiology is learned by seeing cause and effect. CoroSim makes the cause something a learner can change.

- **One model from physiology to fellowship**

  The same heart runs from first-year physiology to the cath lab, so learners build one mental model and keep it

- **Practice of rare, high-stakes events**

  Unlimited low-stakes repetitions of STEMI, shock and heart block, which rotations cannot guarantee

- **Complements existing simulation**

  Comes before mannequin, procedural and ACLS sessions, so lab time goes to hands and teamwork

Fits into the programs you already run

Opens anywhere

Laptop, phone or projector; no install, account or license

Fits current sessions

ECG conference, morning report, flipped pre-work, boot camps, physiology labs

Faculty assign exact cases

Share links open a set patient, tracing or cath case for a lecture, LMS module or exam

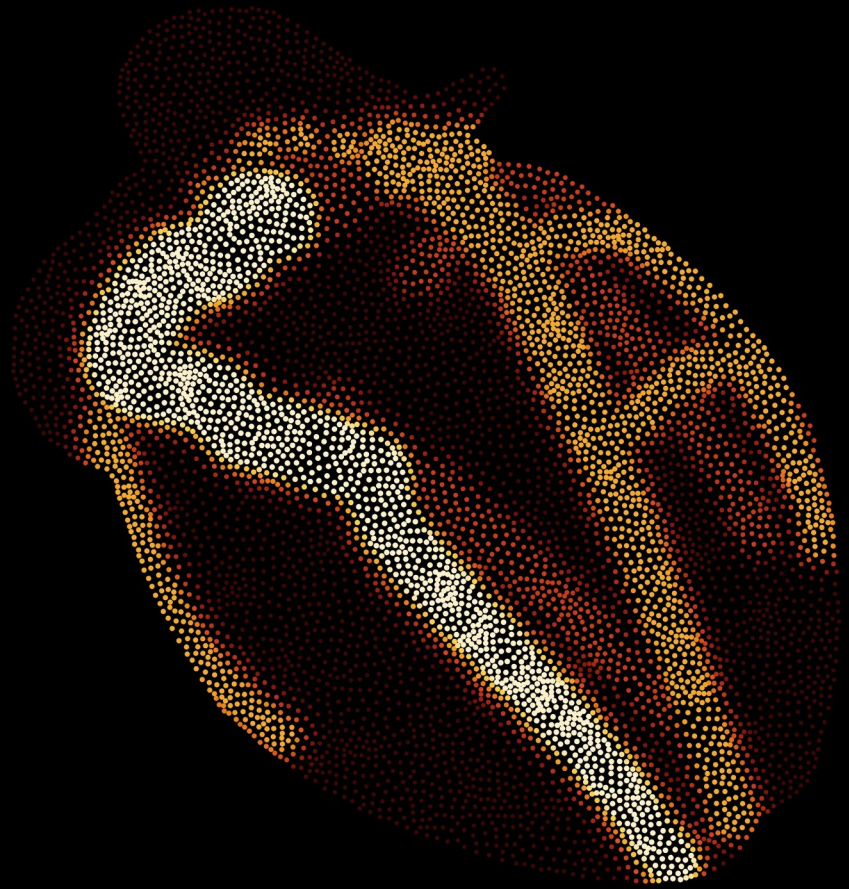
AI tutor on request

Connect AI lets Claude or ChatGPT run Socratic rounds on the learner's patient

A daily habit

One 'Find the culprit' case a day builds spaced, scored practice

Open about its limits: every model constant is tagged literature-backed or assumed, 280+ automated tests run on each change, and it is for education only. Validation against patient data is the next step.



Time is muscle.

Practice is free.

corosim.com

Educational simulation; not for clinical decisions.

REFERENCES

Sources for the problem statement

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